SCALABILITY & FUTURE PLAN

Date	03 October 2026
Team ID	SWTID-2026-8237
Team Members	3
Project Name	Comic Craft - AI Comic Story Creator using Gemini Models

Current System Limitations:

S. No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Single-comic generation without user account management	Users cannot maintain personal comic histories, profiles, or saved libraries.	Medium
2	AI image and story generation can require significant processing resources	Higher concurrent usage may increase response time and resource consumption.	High
3	Current deployment is focused on local execution	Large-scale public access and cloud-based scaling are not yet implemented.	High

Scalability Plan:

S.No	Scalability Aspect	Current State / Proposed Upgrade / Solution
1	User Load	Current system supports local use. Future deployment can use cloud hosting, load balancing, and asynchronous/background AI generation to support larger concurrent user loads.
2	Data Storage	Generated PDFs and panel images are currently stored in project/static directories.

		Future versions can use scalable cloud object storage and a database for user profiles and comic libraries.
3	Performance	Optimize AI generation, cache reusable resources, use background jobs/queues, and scale backend services horizontally to reduce waiting time under higher demand.
4	Security	Use secure environment-variable management for API keys, authentication for user accounts, access controls for saved comics, HTTPS, and secure API handling in a production deployment.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Cloud deployment, scalable backend hosting, and improved AI-generation workflow.	3–6 months	Supports more users and improves availability and scalability.
Phase 3	User accounts, saved comic libraries, comic history, and multi-page story arcs.	6–12 months	Enables personalized long-term use and richer storytelling.
Phase 4	Advanced customization, improved model integration, analytics, and production-grade security.	12+ months	Expands functionality and prepares Comic Craft for broader usage.

Project Basis: Comic Craft uses Fast API for backend routing, Gemini Flash/Pro for structured and detailed story generation, Stable Diffusion for comic-style illustrations, and FPDF for PDF export. The current project documentation identifies future enhancements such as user profiles, comic libraries, and multi-page story arcs.